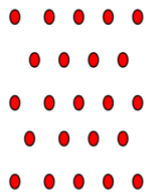


Practice 1: choose 2D bravais lattice

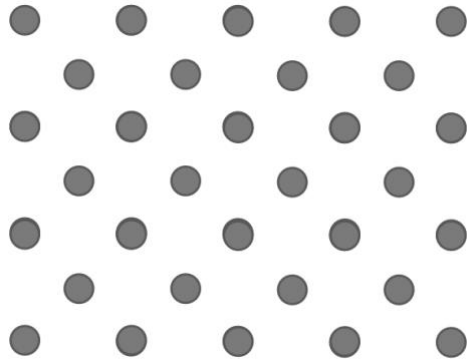
Plot the corresponding Bravais lattice/ unit cell of the following 2D patterns:



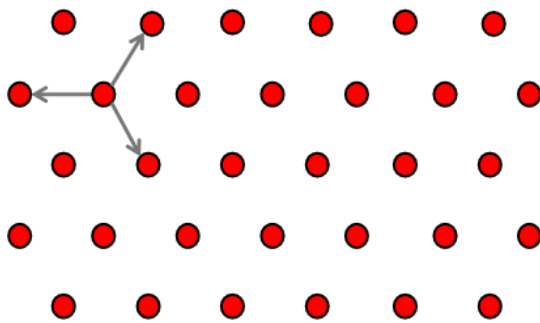
$4mm(L^4P)$



$mm2(L^22P)$



Practice 2: choose the unit cell of 2-D hexagonal lattice

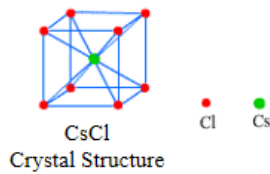
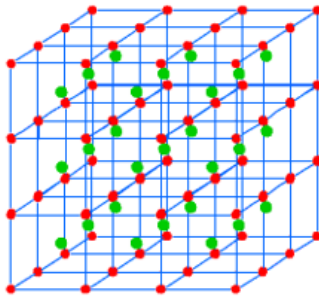


14 bravais lattices

Homework: Please derive 14 bravais lattices

	P	C	I	F
Triclinic		C=P	I=P	F=P
Monoclinic			I=C	F=C
Orthorhombic				
Tetragonal		C=P		
Rhombohedral		Inconsistent with the symmetry of the crystal system	I=R	F=R
Hexagonal			Inconsistent with the condition of the space lattice	
Cubic				

14 bravais lattices

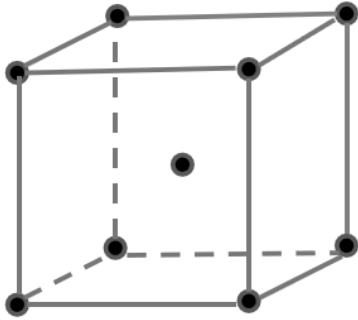


Practice 3: Please point out the lattice type of NaCl and CsCl crystals, respectively.

HOMEWORK

Please classify the 2-D Bravais lattices and point out the properties of their lattice parameters.

PRACTICE4: DENOTATION OF CRYSTALLOGRAPHIC DIRECTIONS

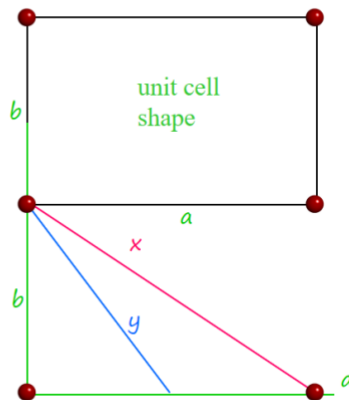


1. Please plot the directions in the left picture, $[001]$, $[101]$, $[110]$, $[1-10]$, $[1-11]$, $[112]$ and $[12-3]$, respectively
2. Draw up all of the directions belong to the family of direction $\langle 111 \rangle$.

PRACTICE5: DENOTATION OF PLANE

$\mathbf{x} = (? \ ? \ ?)$

$\mathbf{y} = (? \ ? \ ?)$



axial ratio = $a:b = 1.60$

PRACTICE5: DENOTATION OF PLANE

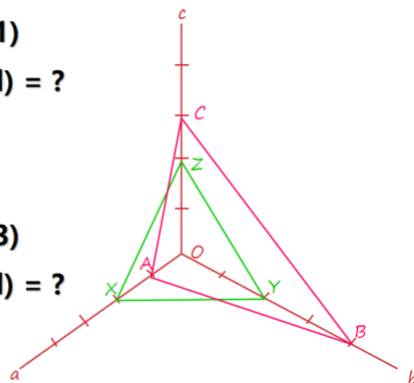
$XYZ = (1 \ 1 \ 1)$

$ABC = (h \ k \ l) = ?$

How about

$XYZ = (1 \ 2 \ 3)$

$ABC = (h \ k \ l) = ?$

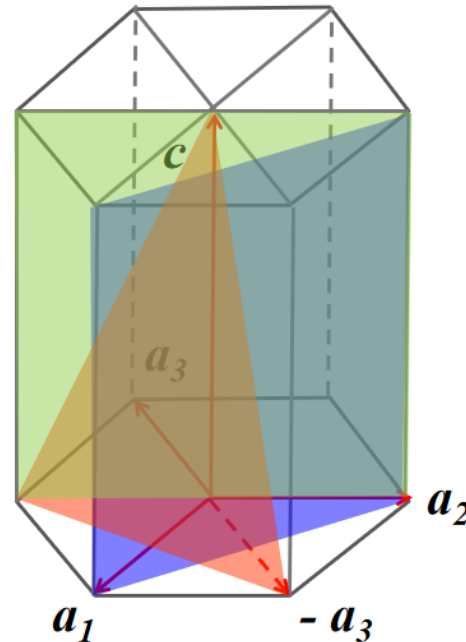


Index of directions and planes in hexagonal crystal system

Practice 8: denotation of planes

Q1. Please give the indices of the three colorful planes in the right picture based on three-axis and four-axis coordinate system, respectively;

Q2. Please plot the planes, (-112) , (-1100) , $(11-20)$, (123) and $(12-33)$ in the hexagonal unit cell.



Index of directions and planes in hexagonal crystal system

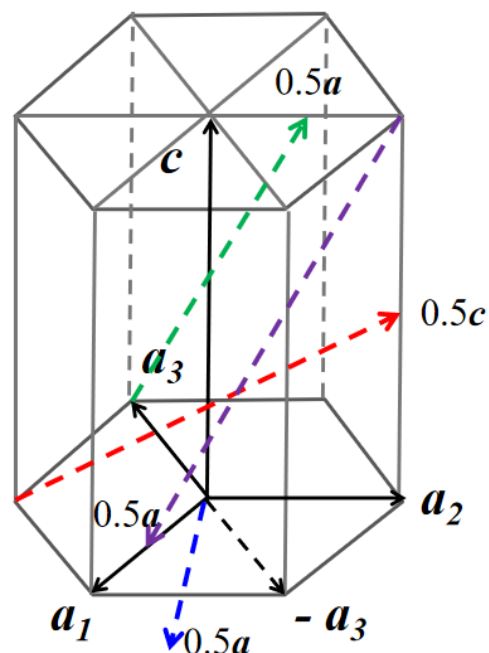
Practice 9:

Q1. Please give the indices of the three colorful directions in the right picture based on the three-axis coordinate system and four-axis coordinate system, respectively;

Q2. Please plot the directions $[101]$, $[11-1]$, $[121]$, $[1-11]$, $[-2111]$ and $[-12-11]$ in the hexagonal lattice.

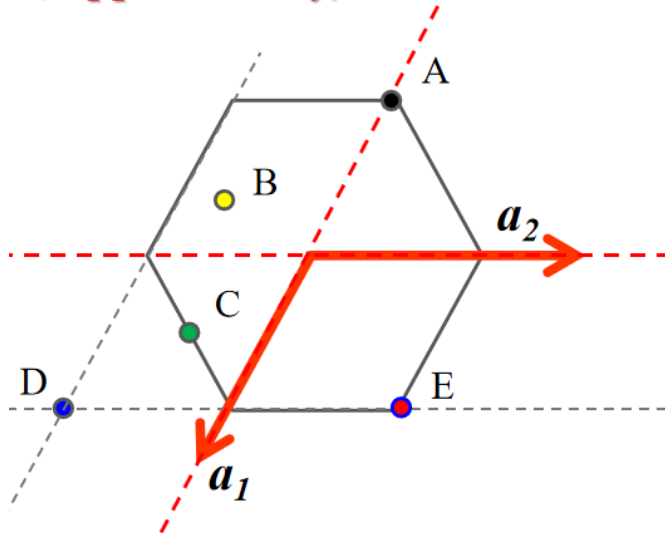
Q3. Point out in Q2 which of them belong to the same group of direction.

$$\begin{aligned} u &= [2U - V]/3; v = [2V - U]/3; \\ t &= -[U + V]/3; w = W \\ \underline{U} &= u - t; \underline{V} = v - t; \underline{W} = w \end{aligned}$$



Index of directions and planes in hexagonal crystal system

How to get the coordinates of a point in a Non - Cartesian coordinate system?
(supplementary)



Practice:

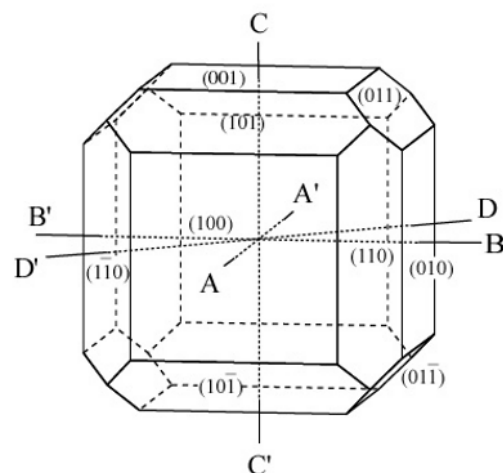
Give the coordinates of the points of A-E.

ZONE & ZONE AXIS

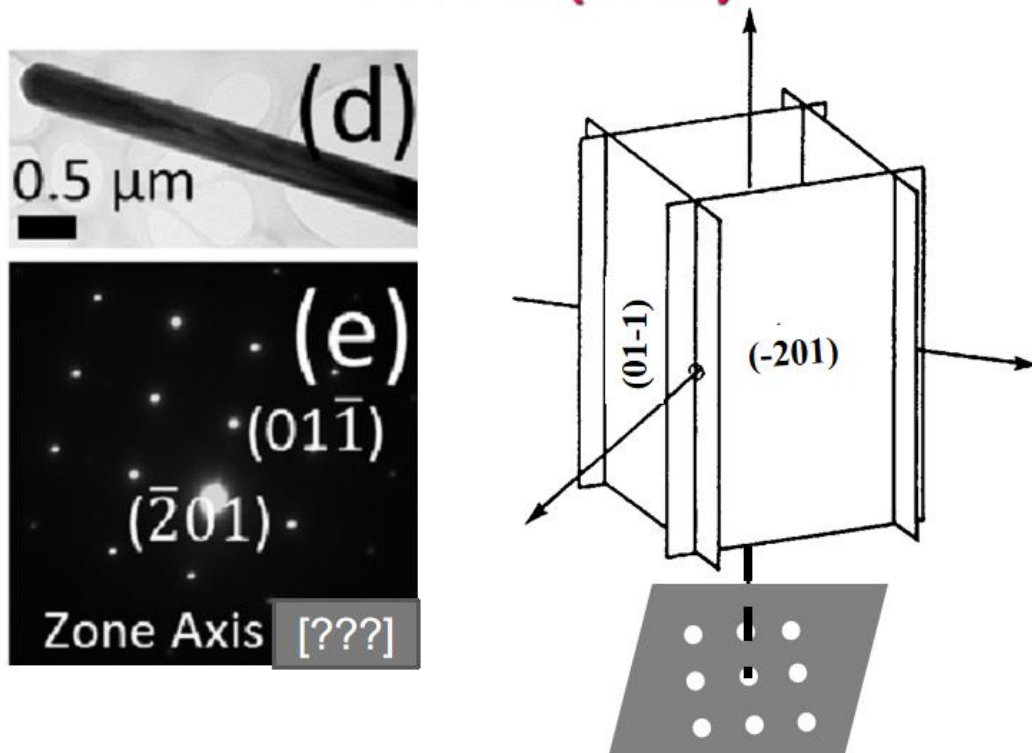
Exercise 10

The right figure shows a cubic crystal with several naked facets.

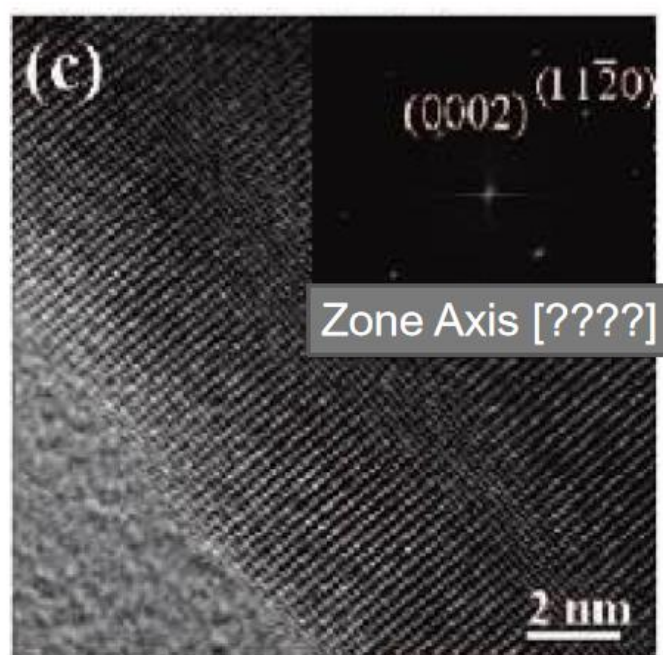
Please give the planes that belong to Zones AA', BB', CC' and DD'.



PRACTICE11: APPL. PHYS. LETT. 100, 103111 (2012)



PRACTICE12: ACS NANO 3 (2009) 53



PRACTICE13: J. PHYS. CHEM. C 115 (2011) 78

Please give the calculation process to show how to get the zone axis in Figure b, c, f, and g.

